

1 Partial Order Relations

1.1 Antisymmetry

No symmetry, **at all**. Let R be a relation on a set A . R is **antisymmetric** iff $\forall a, b \in A, a R b \wedge b R a \rightarrow a = b$.

1.2 Partial Order Relations

Let R be a relation defined on a set A . R is a **partial order relation** if, and only if, R is reflexive, antisymmetric, and transitive.

1.2.1 Example

Let P be the set of all people who have ever lived and define a relation R on P as follows: $\forall r, s \in P, r R s \Leftrightarrow r$ is an ancestor of s or $r = s$. Is R a partial order relation?

Yes. Proof.

R is reflexive Suppose $r \in P$. Then $r = r$, by the definition of R , $r R r$

R is antisymmetric Suppose $r, s \in P, s \in P \wedge r R s \wedge s R r$. We must show that $r = s$. By definition of R , either r is an ancestor of s , or $r = s$, and also, either s is an ancestor of r , or $s = r$. Now it is impossible for both r to be an ancestor of s and for s to be an ancestor of r . Hence one of the conditions must be false, and so $r = s$.

R is transitive Suppose $r, s, t \in P \wedge r R s \wedge s R t$. We must show that $r R t$. By definition of R , either r is an ancestor of s , or $r = s$ and either s is an ancestor of t , or $s = t$.

case 1 In case r is an ancestor of s and s is an ancestor of t , then r is an ancestor of t , and so $r R t$.

case 2 In case r is an ancestor of s and $s = t$, then r is an ancestor of t , and so $r R t$.

⋮

case 3 ⋯

case 4 ⋯

1.2.2 Example

Define a relation r on $\mathbb{Z}$ as follows: $\forall m, n \in \mathbb{Z}, m R n \Leftrightarrow$ every prime factor of m is a prime factor of n .

No, because it's not antisymmetric. Counter example: let $m = 2 \wedge n = 4$. Then $m R n$ because every prime factor of 2 is a prime factor of 4, and $n R m$ because 4 is a prime factor of 4. But $m \neq n$, because $2 \neq 4$.

1.3 Dictionary or Lexicographic

Let A be a set with a partial order relation R , and let S be a set of strings over A . Define a relation $\preceq$ on S as follows:

For any two strings in S , $a_1a_2 \cdots a_m$ and $b_1b_2 \cdots b_n$, where m and n are positive integers,

1. $m \leq n$ and $a_i = b_i$ for all $i = 1, 2, \dots, m$, then

$$a_1a_2 \cdots a_m \preceq b_1b_2 \cdots b_n.$$

2. If for some integer k with $k \leq m, k \leq n$, and $k \geq 1, a_i = b_i$ for all $i = 1, 2, \dots, k-1$ and $a_k \neq b_k$, but $a_k R b_k$ then.

$$a_1a_2 \cdots a_m \preceq b_1b_2 \cdots b_n.$$

3. If ϵ is the null string and s is any string in S , then $\epsilon \preceq s$.

If no strings are related other than by these three conditions, then $\preceq$ is a partial order relation.

1.3.1 Example

Let $A = \{a, b\}$ and suppose A has the partial order relation $R = \{(a, a), (a, b), (b, a), (b, b)\}$. Let S be the set of all strings in a's and b's and let $\preceq$ be the corresponding lexicographic order on S .

C: $\epsilon \preceq aba$? Yes, property #3.

E: $bbab \preceq bbaa$? No, property #2.

F: $ababa \preceq ababaa$? Yes, property #1.